Lecture 31

Taylor series and termwise differentiation

Uses.

Summary 2: Standing fact E includes the logarithm product law for positive arguments.
Summary 8: Product, quotient and chain rules and the elementary derivative table hold.
Summary 11: Finite Taylor polynomials have a Lagrange remainder. The smooth function e^{-1/x^2}, extended by zero, has every derivative zero at zero.
Summary 15: Complex arithmetic, conjugation and modulus obey their algebraic rules and triangle inequality.
Summary 16: The notation $e^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ and its exponential product law are defined for real θ.
Summary 26: The fundamental theorem differentiates accumulation integrals of continuous functions and evaluates definite integrals from primitives.
Summary 29: Absolute convergence implies convergence. Geometric sums and the limits $c^n/n! \to 0$ and $nq^n \to 0$ are proved for real c and $|q| < 1$. Finite grouping preserves established sums. The ratio test proves absolute convergence when its limiting ratio is below 1.
Summary 30: Power series converge uniformly on closed intervals inside their radius. Their sums are continuous and may be integrated termwise there. The bounded-term radius, coefficient-ratio radius formula and integrated-series radius theorem hold.

Objectives. *Uniform convergence alone does not preserve derivatives. For power series, the derivative series has the same radius, which provides the additional control. Taylor remainders then decide whether a function equals its series. We prove the elementary expansions and identify the complex exponential series on the imaginary axis.*

By the end of this lecture you will be able to:

- Prove differentiation of a power series inside its radius and recover its coefficients from derivatives when the radius is positive.
- Justify the exponential, sine and cosine series using finite remainder bounds.
- Derive the logarithmic series and its conditional endpoint value at $x = 1$.
- Split an absolutely convergent complex series into its real, imaginary, even and odd parts.

31.1 The derivative series has the same radius

Suppose a real power series has radius R and sum $F(x) = \sum_{n=0}^{\infty} a_n x^n$ for $|x| < R$. Its *formal derivative series* means the series obtained by differentiating each individual monomial:

$$D(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

At this stage the name makes no assertion that F is differentiable. This is itself a power series, with coefficient $b_m = (m+1)a_{m+1}$ at x^m for $m \geq 0$. Let its radius be R_D, supplied by Lecture 30.

Theorem 31.1 (Termwise differentiation). *The derivative series has radius $R_D = R$. For every $|x| < R$, the sum F is differentiable and $F'(x) = D(x)$.*

Proof. First compare the radii. For $0 < r < \rho < R$, the original series converges at ρ, so its terms are bounded by some M: $|a_n \rho^n| \leq M$ for all n. For $n \geq 1$,

$$|n a_n r^{n-1}| \leq \frac{M}{r} n \left(\frac{r}{\rho}\right)^n \longrightarrow 0$$

by the proved growth limit of Lecture 29. The derivative-series terms at r are therefore bounded, so r belongs to its bounded-term set from Lecture 30. This holds for every $0 < r < R$, and hence $R_D \geq R$. If $R = 0$ the inequality is automatic; if $R = \infty$ it already identifies both radii.

If $R < \infty$ and $R_D > R$, choose $R < \rho < R_D$. Convergence of the derivative series at this positive point bounds $n|a_n|\rho^{n-1} \leq K$ for $n \geq 1$. It follows that $|a_n \rho^n| \leq K\rho/n \leq K\rho$. Including the finite constant term, the original terms at ρ are bounded. This contradicts $\rho > R$, the supremum of the original bounded-term set. Thus the radii are equal.

Now fix $|x| < r < R$. The derivative series converges uniformly on $[-r, r]$ and has a continuous sum D, by Lecture 30. Integrate it termwise between 0 and x:

$$\int_0^x D(t)\,\mathrm{d}t = \sum_{n=1}^{\infty} n a_n \frac{x^n}{n} = F(x) - a_0.$$

For negative x, use the endpoint-reversal convention. The fundamental theorem makes the left side differentiable with derivative $D(x)$. The identity therefore proves that F is differentiable and $F' = D$, without presupposing that derivative. ∎

Only interior points are covered. Convergence of an original series at an endpoint does not give a derivative there by this theorem.

Example 31.2 (Differentiating a known weighted sum). Lecture 30's integrated-geometric example gives $\sum_{n=0}^{\infty}(n+1)x^n = 1/(1-x)^2$ for $|x| < 1$. Multiply by x and reindex to write

$$H(x) = \sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}.$$

The proved derivative rule now gives $xH'(x) = \sum_{n=1}^{\infty} n^2 x^n$. Ordinary differentiation of the displayed rational function yields

$$\sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3} \qquad (|x| < 1).$$

The same-radius theorem controls the infinite derivative operation; the quotient rule computes the resulting finite expression.

Example 31.3 (Uniform convergence does not supply an endpoint derivative)**.** Consider $F(x) = \sum_{n=1}^{\infty} x^n/n^2$ on $[0,1]$. The majorant $1/n^2$ gives uniform convergence and continuity on this closed interval. Its radius is 1 by the coefficient ratio test. The formal derivative at $x = 1$ would be the divergent harmonic series. We can also show directly that the function has no finite left derivative there.

For $0 < h < 1$, subtraction of convergent series gives a sum of nonnegative terms:

$$\frac{F(1) - F(1-h)}{h} = \sum_{n=1}^{\infty} \frac{1-(1-h)^n}{hn^2} \geq \sum_{n=1}^{M} \frac{1-(1-h)^n}{hn^2}$$

for every fixed integer $M \geq 1$. As $h \downarrow 0$, each of the finitely many terms on the right tends to $1/n$, by the polynomial derivative at 1. Given any $L > 0$, choose M whose harmonic partial sum exceeds $2L$. For all sufficiently small positive h, the right side then exceeds L. Thus the left-hand difference quotient tends to $+\infty$. The endpoint is outside the differentiation theorem even though the original series is uniform on the entire closed interval.

31.2 Taylor coefficients and the remainder condition

Let F be a power-series sum with radius $R > 0$. Repeated application of Theorem 31.1 preserves the radius at every finite derivative order. Each derivative series has a continuous sum inside that radius. Thus a power-series sum is C^∞ there: it has continuous derivatives of every order. For each integer $k \geq 0$, with $F^{(0)} = F$, repeated differentiation gives

$$F^{(k)}(x) = \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} a_n x^{n-k}, \qquad F^{(k)}(0) = k! a_k.$$

The first formula follows by induction, multiplying by the next exponent at each step. At zero only the term $n = k$ survives, giving the second. Consequently any power-series representation about zero has coefficients $a_k = F^{(k)}(0)/k!$. They are unique. Replacing x by $x - a$ gives the same conclusion at any centre a.

If a function f has derivatives of every order near a, its *Taylor series about* a is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

The series at $a = 0$ is also called its *Maclaurin series*. The function is *analytic at* a if this series equals $f(x)$ on some open interval $|x - a| < r$ with $r > 0$. Every power-series sum with positive radius is analytic at its centre by the coefficient formula above. A smooth function need not be analytic.

Recall the smooth function of Lecture 11, given by $F(0) = 0$ and $F(x) = e^{-1/x^2}$ for $x \neq 0$. That lecture proves $F^{(n)}(0) = 0$ for every n. Its Taylor series is therefore the zero series, convergent for every real x. But $F(x) > 0$ for $x \neq 0$, so no neighbourhood of zero has equality. The function is not analytic at zero. Merely forming a convergent series from the derivatives does not establish a representation of the function.

For a function with Taylor polynomial T_N about a, write its finite remainder as $R_N(x) = f(x) - T_N(x)$. At a fixed point, the Taylor series represents f exactly when $R_N(x) \to 0$. This is the partial-sum definition of series convergence. The Lagrange remainder from Lecture 11 supplies a way to prove that limit.

31.3 Exponential, sine and cosine series

Theorem 31.4 (Elementary Taylor representations). *For every real x, the following absolutely convergent series equal their functions:*

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \tag{31.1}$$

$$\sin x = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{(2m+1)!}, \tag{31.2}$$

$$\cos x = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{(2m)!}. \tag{31.3}$$

Proof. Every exponential derivative is e^x. Between zero and a fixed x, its absolute value is at most $e^{|x|}$. Taylor's formula through degree N therefore gives

$$\left| e^x - \sum_{n=0}^{N} \frac{x^n}{n!} \right| \le e^{|x|} \frac{|x|^{N+1}}{(N+1)!} \longrightarrow 0,$$

using Lecture 29's factorial growth limit with the fixed real number $|x|$. This proves the exponential representation at every real x.

The sine and cosine derivatives cycle through the following values at zero:

Derivative order modulo 4	0	1	2	3
Sine	0	1	0	−1
Cosine	1	0	−1	0

Every derivative has absolute value at most 1 on the real line. The Taylor remainder through degree N is therefore at most $|x|^{N+1}/(N+1)!$, which again tends to zero. The table supplies the coefficients in Eqs. (31.2) and (31.3); the zero coefficients leave only the stated odd and even powers. Their finite Taylor polynomials are subsequences of the full degree sequence, so the same limit identifies the displayed sums.

Finally, for $x \neq 0$, the positive series $\sum |x|^n/n!$ converges by Lecture 29's ratio test. At $x = 0$, it has only the constant term 1. The absolute sine and cosine terms are its nonnegative subseries, whose partial sums are bounded by its sum. Both converge by nonnegative comparison. This proves absolute convergence of all three representations. ∎

The same remainder argument gives uniform quantitative control on bounded intervals. Fix $A > 0$. For every $|x| \le A$, the exponential error is at most $e^A A^{N+1}/(N+1)!$, and each trigonometric error is at most $A^{N+1}/(N+1)!$. These bounds tend to zero with one choice of N for the whole interval. Thus the proved representations converge uniformly on every $[-A, A]$, as the radius theorem also predicts. They do not assert uniform approximation over the whole real line. For example, each nonconstant Taylor polynomial of sine is unbounded on $\mathbb{R}$, while sine is bounded; its global error cannot be uniformly small. The domain matters even for functions with infinite convergence radius.

Figure 31.1 compares sine with two finite Taylor polynomials. The remainder theorem controls a polynomial at a chosen point; a picture on a finite range is not a replacement for that estimate.

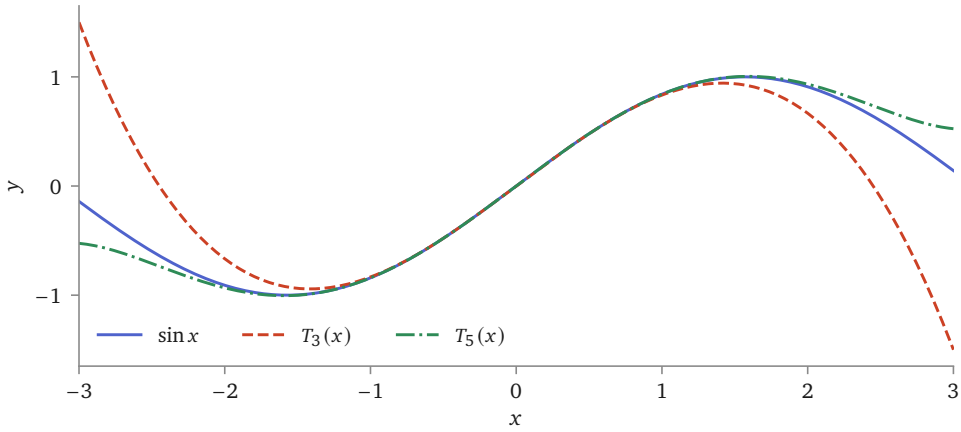


Figure 31.1: Sine and its polynomials $T_3(x) = x - x^3/6$ and $T_5(x) = x - x^3/6 + x^5/120$ on $[-3, 3]$. The remainder bound tends to zero as degree increases at any fixed point.

Example 31.5 (A finite error bound for sine). Approximate $\sin(2/5)$ by $P = x - x^3/6 + x^5/120$ at $x = 2/5$. This is also the degree-6 Taylor polynomial, since its degree-6 coefficient is zero. The degree-6 Lagrange bound therefore gives

$$P = \frac{18254}{46875}, \qquad |\sin(2/5) - P| \le \frac{(2/5)^7}{7!} = \frac{8}{24609375} < \frac{1}{3000000}.$$

The derivative bound is 1 at every possible remainder point. No numerical measurement of the actual error was used to certify it. Using a degree-5 remainder bound would also be valid, but would give a larger bound because it discards the vanishing next Taylor coefficient.

31.4 The logarithm and its endpoint

For $|t| < 1$, the geometric identity gives $1/(1+t) = \sum_{n=0}^{\infty} (-1)^n t^n$. On each $[-r, r]$ with $r < 1$, the terms have convergent majorant r^n, so the series is uniform there. Lecture 30 permits integration from zero to any interior x and gives

$$\ln(1+x) = \int_0^x \frac{dt}{1+t} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} \qquad (-1 < x < 1). \tag{31.4}$$

The primitive is the real logarithm because $1 + t > 0$ on the integration interval. For negative x, the reversed endpoints give the same formula. The coefficient ratio has limit 1, so this power series has radius 1. The function itself is defined for every $x > -1$, but this particular series need not converge at points outside its radius.

At $x = 1$, the geometric integrand series does not converge, so the preceding uniform-integration argument cannot be applied there directly. Instead apply the finite Taylor theorem to $f(x) = \ln(1+x)$. For $n \ge 1$, induction from the logarithm derivative gives

$$f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}.$$

At $x = 1$, Taylor's formula through degree $N \geq 1$ has remainder

$$\ln 2 - \sum_{n=1}^{N} \frac{(-1)^{n-1}}{n} = \frac{(-1)^N}{(N+1)(1+\xi)^{N+1}} \quad \text{for some } \xi \in (0,1).$$

Its absolute value is at most $1/(N+1)$ and tends to zero. Thus the conditional endpoint value is $\ln 2 = \sum_{n=1}^{\infty}(-1)^{n-1}/n$. At $x = -1$, the terms become $-1/n$ and the series diverges. Consequently Eq. (31.4) extends to $-1 < x \leq 1$, with absolute convergence for $|x| < 1$ and conditional convergence at 1. This identifies Lecture 30's uniform alternating sum on $[0, 1]$.

Example 31.6 (A different centre extends the usable interval)**.** The failure of the Maclaurin series at $x > 1$ does not mean that $\ln(1+x)$ fails to be analytic there. Fix any centre $a > -1$ and put $d = 1 + a > 0$. For $|x - a| < d$, the number $y = (x-a)/d$ satisfies $|y| < 1$ and $1 + x = d(1+y) > 0$. The logarithm product law and the proved series give

$$\ln(1+x) = \ln d + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(x-a)^n}{nd^n} \qquad (|x-a| < d).$$

Its coefficients agree with $f^{(n)}(a)/n!$ from the derivative formula. It is therefore the Taylor series at a, with radius d by the coefficient ratio test. Every centre in the logarithm's domain has a positive radius, proving analyticity at each such point.

For instance, at centre $a = 2$ the radius is 3. This expansion can evaluate the function at $x = 3$, where its original Maclaurin series diverges. The new variable is $y = 1/3$, and the result is

$$\ln 4 = \ln 3 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n3^n}.$$

At the new upper endpoint $x = 2a + 1$, the series has value $\ln d + \ln 2$ by the already proved conditional endpoint sum. The lower endpoint is always $x = -1$, where the series diverges and the logarithm is undefined. The centre determines the useful expansion interval.

31.5 Complex series and the imaginary exponential

A complex sequence z_n converges to z if $|z_n - z| \to 0$. This is equivalent to convergence of both real components. Indeed, each component error is at most the modulus error, while

$$|z_n - z| \leq |\operatorname{Re}(z_n - z)| + |\operatorname{Im}(z_n - z)|.$$

A *complex series* converges when its complex partial sums converge; it *converges absolutely* when the real series $\sum |z_n|$ converges. For an absolutely convergent complex series, the real and imaginary component series are absolutely convergent because $|\operatorname{Re} z_n| \leq |z_n|$ and $|\operatorname{Im} z_n| \leq |z_n|$. Lecture 29 gives their real limits, and the component equivalence gives the complex limit. Absolute convergence therefore implies complex convergence. Passing the finite triangle inequality to the limit also bounds its modulus by $\sum |z_n|$.

For such a series indexed from zero, both the even and odd subseries converge absolutely: their finite absolute sums are bounded by the full absolute sum. On finite paired cutoffs,

$$\sum_{n=0}^{2N+1} z_n = \sum_{m=0}^{N} z_{2m} + \sum_{m=0}^{N} z_{2m+1}.$$

Every term on this line has a limit, proving that the full sum is the sum of its even and odd subseries. This conclusion follows from finite identities and absolute convergence, rather than arbitrary formal rearrangement.

Now fix $\theta \in \mathbb{R}$ and take $z_n = (\mathrm{i}\theta)^n/n!$. The constant term is 1, including when $\theta = 0$. Their moduli are $|\theta|^n/n!$, whose series converges by the real absolute-convergence result just proved. At zero it is a finite constant series. Since $\mathrm{i}^{2m} = (-1)^m$ and $\mathrm{i}^{2m+1} = \mathrm{i}(-1)^m$, even/odd separation gives

$$\begin{aligned}\sum_{n=0}^{\infty} \frac{(\mathrm{i}\theta)^n}{n!} &= \sum_{m=0}^{\infty} \frac{(-1)^m\theta^{2m}}{(2m)!} + \mathrm{i}\sum_{m=0}^{\infty} \frac{(-1)^m\theta^{2m+1}}{(2m+1)!} \\ &= \cos\theta + \mathrm{i}\sin\theta = e^{\mathrm{i}\theta}.\end{aligned}$$

The sine and cosine representations were proved above, and the final notation is exactly Lecture 16's definition. Thus that geometric exponential agrees with the factorial series on the imaginary axis. Lecture 34 proves the series representation for every complex argument using a uniqueness result from Lecture 33. No complex-variable differentiation theorem is assumed here.

Summary 31.

- Theorem 31.1: the formal derivative has the same radius; inside it, termwise differentiation is valid. For $R > 0$, repeated differentiation gives $a_n = F^{(n)}(0)/n!$ and unique coefficients.
- A smooth function need not equal its Taylor series. Equality at a point is equivalent to the finite Taylor remainder tending to zero there.
- Theorem 31.4: exponential, sine and cosine equal their absolutely convergent Taylor series for every real argument, by the factorial remainder bound.
- Equation (31.4): the logarithmic series holds for $-1 < x \leq 1$. The value at 1 is proved by a finite Taylor remainder; the series diverges at $x = -1$.
- Absolute complex convergence permits component and even/odd separation. The exponential factorial series equals $e^{\mathrm{i}\theta}$ for every real θ.

Sheet 31 retains the binomial-series sketch and the finite Gaussian approximation, alongside transfer tasks for sums and coefficients. Lecture 32 applies differentiation and integration to motion along paths.

References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§11.9–11.11. K. F. Riley, M. P. Hobson and S. J. Bence, *Mathematical Methods for Physics and Engineering*, 3rd ed., Cambridge University Press, 2006, §§4.5–4.6.